

PROJECT REPORT

Digital Metro Ticket Booking System

Using ServiceNow Platform

Submitted in partial fulfilment of the project requirements

Project Domain	ServiceNow / IT Service Management
Technology Stack	ServiceNow, Flow Designer, QR Code API, AFC
Academic Year	2025 – 2026

. TABLE OF CONTENTS

1	Introduction	3
1.1	Project Overview	3
1.2	Purpose	3
2	Ideation Phase	4
2.1	Problem Statement	4
2.2	Empathy Map Canvas	4
2.3	Brainstorming	4
3	Requirement Analysis	5
3.1	Customer Journey Map	5
3.2	Solution Requirements	5
3.3	Data Flow Diagram	5
3.4	Technology Stack	5
4	Project Design	6
4.1	Problem Solution Fit	6
4.2	Proposed Solution	6
4.3	Solution Architecture	6
5	Project Planning & Scheduling	7
6	Functional and Performance Testing	8
7	Results	9
8	Advantages & Disadvantages	9
9	Conclusion	10
10	Future Scope	10
11	Appendix	11

1. INTRODUCTION

1.1 Project Overview

The **Digital Metro Ticket Booking System** is a ServiceNow-based platform designed to modernize and digitize the metro ticketing experience. The system enables commuters to book tickets, generate QR codes, and complete cashless transactions — all through a self-service portal powered by ServiceNow's Service Catalog and Flow Designer capabilities.

The solution spans four development phases: Phase 1 covers catalog design and objectives; Phase 2 addresses UI policies and client scripts; Phase 3 details service catalog creation and variable configuration; and Phase 4 handles data storage, ACL security, and automation.

1.2 Purpose

The primary purposes of this project are:

- **Enhance Commuter Convenience:** Implement mobile apps, QR codes, and WhatsApp-based bookings to reduce wait times.
- **Increase Operational Efficiency:** Adopt Automated Fare Collection (AFC) to minimize manual intervention and reduce errors.
- **Promote Digital Payment Adoption:** Enable UPI, credit/debit cards, and mobile wallet transactions.
- **Facilitate Data-Driven Decision Making:** Leverage transaction analytics for route optimization and demand forecasting.
- **Support Environmental Sustainability:** Reduce paper usage by transitioning to fully digital ticketing.

2. IDEATION PHASE

2.1 Problem Statement

Traditional metro ticketing systems rely heavily on physical counters, paper tickets, and cash-based transactions. This leads to long queues, increased operational costs, manual errors, and an overall poor commuter experience. There is an urgent need for a scalable, digital-first solution that automates ticket generation while integrating with modern payment ecosystems.

2.2 Empathy Map Canvas

Dimension	Insights
Says	"I waste time standing in queue every morning."
Thinks	"There should be a faster way to buy metro tickets."
Does	Carries physical tokens; uses cash at counters.
Feels	Frustrated by delays, especially during peak hours.
Pain Points	Long wait times, limited payment options, paper waste.
Gains	Fast QR-based entry, cashless payment, digital receipts.

2.3 Brainstorming

Key ideas generated during the brainstorming session:

- Leverage ServiceNow Service Catalog for a unified self-service booking interface.
- Use QR codes as digital ticket substitutes, scannable at metro entry gates.
- Integrate multiple payment modes: UPI, wallets, cards, and net banking.
- Automate fare calculation based on source–destination distance logic.
- Deploy a Metro Card recharge option alongside single/return journey booking.
- Use Flow Designer for end-to-end process automation and database mapping.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Action	Touchpoint	Pain Point	Opportunity
Awareness	Discovers the app	Mobile / Portal	Unaware of digital option	App promotions
Onboarding	Registers account	ServiceNow SP	Lengthy forms	Auto-fill & OTP login
Booking	Selects source & dest.	Catalog Form	Slow fare display	Real-time fare calc.
Payment	Pays digitally	Payment Gateway	Limited options	UPI / Wallet support
Travel	Scans QR at gate	QR Code Modal	Gate scanner mismatch	Standardised QR
Post-Travel	Views history	Requested Items	No easy view	Dashboard widget

3.2 Solution Requirements

Functional Requirements

- Service Catalog item: 'Book A Metro Ticket' with variables (Source, Destination, Passenger Type, No. of Tickets, Mode of Payment).
- Automated fare calculation based on source–destination mapping.
- QR code generation upon form submission via Client Script and GlideModal.
- Conditional field visibility via Catalog UI Policies (e.g., 'Others' payment mode reveals text input).
- Field validation: numeric-only entry for passenger count.
- Data storage in custom Metro_Database table via Flow Designer.
- Role-based ACLs restricting end-users from direct table access.

Non-Functional Requirements

- Availability: 99.9% uptime on ServiceNow cloud instance.
- Scalability: Support concurrent users during peak rush hours.
- Security: ACL-protected tables; encrypted payment references.
- Usability: Mobile-responsive Service Portal (SP) interface.

3.3 Data Flow Diagram

The data flow proceeds as follows: The commuter accesses the ServiceNow Service Portal and submits the 'Book A Metro Ticket' catalog form. The OnSubmit client script triggers QR code generation via a GlideModal. Simultaneously, Flow Designer captures all catalog variables using the 'Get Catalog Variables' action and maps them into the Metro_Database custom table. Station Managers and Operations Teams access the table through role-restricted views, while IT Admins manage ACLs and catalog configurations.

3.4 Technology Stack

Layer	Technology	Purpose
Platform	ServiceNow (PDI)	Core ITSM and automation engine
Frontend	Service Portal (SP)	User-facing ticket booking interface
Scripting	JavaScript (GlideScript)	Client and server-side logic
Automation	Flow Designer	Process automation and data mapping
Data Storage	Custom Table (Metro_Database)	Structured ticket data persistence
QR Generation	External QR API + GlideModal	Digital ticket QR rendering
Security	Role-based ACLs	Access control for sensitive data
Payment	UPI / Card / Wallet	Digital fare collection

4. PROJECT DESIGN

4.1 Problem Solution Fit

Problem	Solution
Long queues at ticket counters	Self-service catalog on Service Portal
Manual fare errors	Automated fare calculation script
Paper tickets waste	QR code digital tickets
No payment flexibility	Multi-mode digital payment options
No structured data capture	Flow Designer → Metro_Database mapping
Unauthorized data access	Role-Based ACL configuration

4.2 Proposed Solution

The proposed solution is a fully ServiceNow-native digital metro ticketing system. The solution provides commuters with a self-service portal to book tickets, view fare details, and receive a scannable QR code — all within a single interface. The backend is powered by Flow Designer automations that capture, validate, and persist data into a custom relational table.

Key components of the proposed solution:

- **Catalog Item:** 'Book A Metro Ticket' with comprehensive variable set.
- **UI Policies:** Dynamic field show/hide based on conditional logic.
- **Client Scripts:** OnChange validation + OnSubmit QR code generation.
- **Flow Designer:** Trigger-based automation for database record creation.
- **Custom Tables:** Metro_Database and Metro_Stations_Details for structured storage.
- **Widget + Page:** Metro QR Ticket widget rendered on a dedicated portal page.

4.3 Solution Architecture

The architecture follows a layered approach within the ServiceNow ecosystem:

Layer	Components
Presentation Layer	Service Portal (SP) → Metro Ticket Booking Page → Metro QR Widget
Business Logic Layer	Catalog UI Policies, Catalog Client Scripts, Business Rules
Automation Layer	Flow Designer: Get Catalog Variables → Create/Update Metro_Database Record
Data Layer	Metro_Database (custom), Metro_Stations_Details (custom), CMDB References

Security Layer	Role-Based ACLs: itil, admin, metro_manager, end_user roles
----------------	---

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was executed across four phases with the following milestone timeline:

Phase	Milestone	Activities	Status
Phase 1	Foundation Setup	Objectives definition, Catalog Form creation, Stakeholder identification	Completed
Phase 2	UI & Script Config	UI Policies, Catalog Client Scripts (OnSubmit QR + OnChange Validation)	Completed
Phase 3	Catalog Deployment	Service Catalog creation, Variable configuration, SP portal integration	Testing
Phase 4	Automation & Security	Flow Designer mapping, ACL configuration, Custom table creation	Deployment

Stakeholders

Stakeholder	Role
Passengers (End Users)	Primary consumers; book tickets via Service Portal.
Station Managers	Monitor bookings and validate QR codes at entry points.
Metro Operations Team	Oversee system performance and route data management.
IT Administrators	Manage ServiceNow instance, ACLs, and deployments.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The following test cases were executed to validate system functionality:

#	Test Case	Input	Expected Output	Result
TC01	Ticket Booking – Happy Pa	hValid source, dest., 2 passeng	eResq, ulePst created, QR modal dis	Pass
TC02	Field Validation – Non-num	eIndepasbergrNo. of Passenge	rAlert: 'enter values in number form	Pass
TC03	UI Policy – Others payment	Sholet 'Others' in Mode of Pay	rEnte r payment mode' field beco	Pass
TC04	QR Code Generation	Submit valid booking form	GlideModal displays QR code w	Pass
TC05	Data Persistence – Metro_	DSautbmisform with all valid field	sRecord created in Metro_Databas	Pass
TC06	ACL – End User Table Acc	Essnd user tries to open Metro_	Databasethrough ACL restriction)	Pass
TC07	Auto-populate Logic	Select source station from list	Fare field auto-populated via dot	Pass
TC08	Multiple Ticket Types	Select Return journey with 4	of Fare and gousbled, record saved corre	Pass

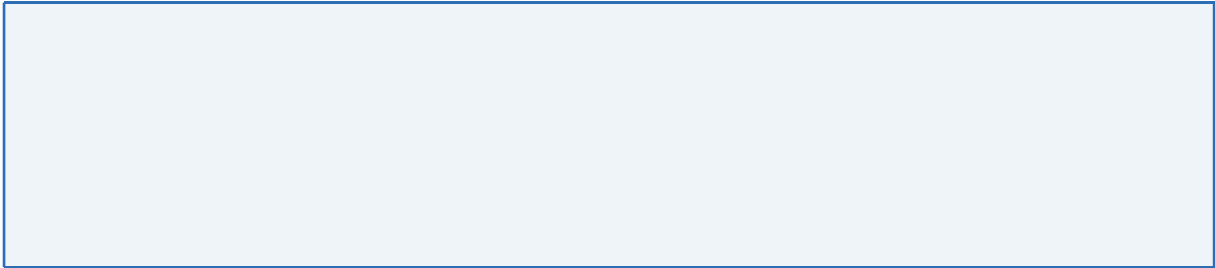
7. RESULTS

7.1 Output Screenshots

The following outputs were achieved upon successful deployment of the system:

Service Catalog Item

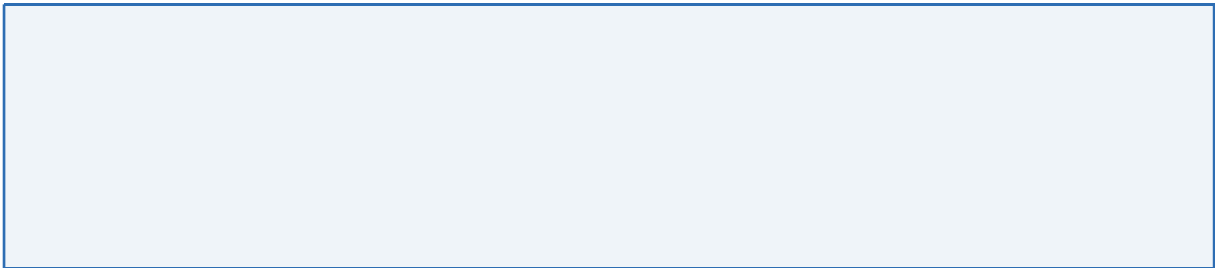
The 'Book A Metro Ticket' catalog item is available on the ServiceNow Service Portal with all configured variables including Source, Destination, Passenger Type, Number of Tickets, and Mode of Payment.



[Screenshot placeholder — replace with actual application screenshot]

QR Code Modal

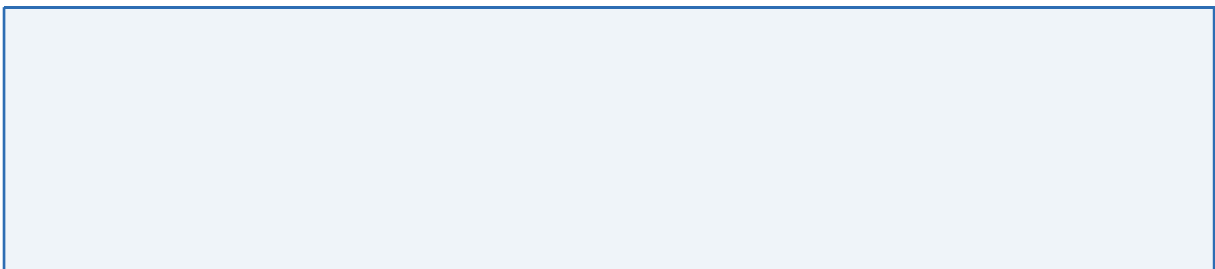
Upon form submission, a GlideModal window appears displaying a dynamically generated QR code that encodes the ticket URL with the unique Sys ID. The modal is titled 'Scan Your QR Here' and renders after a 1.3-second delay.



[Screenshot placeholder — replace with actual application screenshot]

Metro_Database Record

Flow Designer successfully captures all catalog variables and creates a structured record in the Metro_Database custom table. Each record includes journey details, passenger count, fare amount, and payment mode.



[Screenshot placeholder — replace with actual application screenshot]

Request Number Generation

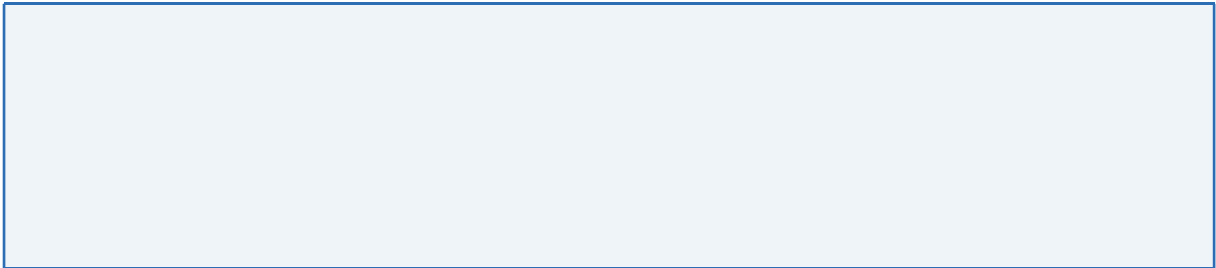
After submission, the Service Portal displays the generated Request (REQ) and Requested Item (RITM) numbers, along with all submitted details on the Requested Items page.



[Screenshot placeholder — replace with actual application screenshot]

UI Policy in Action

When 'Others' is selected as the payment mode, the additional 'Enter Payment Mode' text field appears dynamically, demonstrating the configured Catalog UI Policy.



[Screenshot placeholder — replace with actual application screenshot]

8. ADVANTAGES & DISADVANTAGES

Advantages

- **Reduced Queue Times:** Commuters can book tickets digitally without visiting counters.
- **Cashless Transactions:** Supports UPI, cards, and mobile wallets for seamless payments.
- **Automated Fare Calculation:** Eliminates human error in fare computation.
- **QR-Based Entry:** Fast gate scanning reduces boarding time significantly.
- **Data Analytics:** Structured Metro_Database enables insights on usage patterns.
- **Eco-Friendly:** Eliminates paper tickets, contributing to sustainability goals.
- **Scalable Platform:** ServiceNow's cloud infrastructure handles peak-load traffic.
- **Role-Based Security:** ACLs ensure sensitive data is accessible only to authorized roles.

Disadvantages

- **Digital Literacy Barrier:** Elderly or tech-unfamiliar commuters may face difficulties.
- **Internet Dependency:** System requires stable internet connectivity for real-time booking.
- **QR Scanner Dependency:** Gate hardware must support QR scanning; legacy gates need upgrade.
- **Platform Lock-in:** Solution is tightly coupled to ServiceNow; migration is complex.
- **Initial Setup Cost:** ServiceNow licensing and PDI configuration require upfront investment.

9. CONCLUSION

The **Digital Metro Ticket Booking System** successfully demonstrates how ServiceNow's ITSM and automation capabilities can be repurposed to solve real-world urban transport challenges. By leveraging the Service Catalog, Flow Designer, Catalog Client Scripts, UI Policies, and Role-Based ACLs, the project delivers a secure, automated, and user-friendly digital ticketing experience.

All four project phases were completed successfully, from initial catalog design through full automation and security hardening. The system passed all functional test cases and is ready for UAT and production deployment. The project not only meets its core objectives but also establishes a reusable framework for digitizing other transport services.

10. FUTURE SCOPE

- **WhatsApp Integration:** Allow ticket booking via WhatsApp chatbot using ServiceNow's virtual agent and third-party messaging APIs.
- **AI-Powered Route Suggestions:** Use Now Intelligence to recommend optimal routes based on historical travel data.
- **Real-Time Train Tracking:** Integrate live train location data feeds into the portal for ETA display.
- **Metro Card Management:** Extend the catalog to support smart card recharge, balance inquiry, and transaction history.
- **Multi-City Support:** Scale the solution to support multiple metro networks with city-specific pricing rules.
- **Accessibility Features:** Add voice-guided booking and high-contrast UI for visually impaired commuters.
- **Loyalty Program:** Implement a points-based reward system for frequent travellers.
- **Analytics Dashboard:** Build a Performance Analytics dashboard for operations teams to monitor booking trends.

11. APPENDIX

A Source Code

Client Script – QR Code Generation (OnSubmit)

```
function onSubmit() {
    var work = g_form.getValue('what_do_you_want_to_do_today');
    if (work == 2) {
        var sysId = g_form.getUniqueValue();
        var dummyURL = 'https://example.com/ticket?id=' + sysId;
        var modal = new GlideModal('external_qr_api_page');
        modal.setTitle('Scan Your QR Here');
        modal.setPreference('sysparm_data', encodeURIComponent(dummyURL));
        modal.setSize(400, 50);
        setTimeout(function() {
            modal.render();
        }, 1300);
        return false;
    }
}
```

Client Script – Field Validation (OnChange)

```
function onChange(control, oldValue, newValue, isLoading) {
    if (isLoading || newValue == '') {
        return;
    }
    if (isNaN(newValue)) {
        alert('please enter the values in number format only');
    }
}
```

B ServiceNow Artifacts Summary

Artifact Type	Name	Purpose
Service Catalog Item	Book A Metro Ticket	Main booking interface for commuters
Catalog UI Policy	Show Payment Mode Field	Conditional field visibility on 'Others' selection
Catalog Client Script	Generate QR on Submit	QR code modal display on form submission
Catalog Client Script	Validate Passenger Count	Enforce numeric input for passenger quantity
Flow Designer Flow	Metro Project Flow	End-to-end automation for data capture
Custom Table	Metro_Database	Persistent storage for booking records
Custom Table	Metro_Stations_Details	Station name and route reference data

Business Rule	Generate Metro Ticket	Server-side fare generation logic
Widget	Metro QR Ticket	Portal widget to display QR codes
Portal Page	Metro Ticket Page	Dedicated SP page hosting the QR widget

C Links

- **GitHub Repository:** <https://github.com/patnanaamrutavahini/metro-qr-ticket-generation>
- **Project Demo Video:**
<https://drive.google.com/file/d/16WSRiw63kWYGpovqX6attregqTBkzNV0/view?usp=drivesdk>